Answer key: Lesson 7, Make Your Own Graph

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the tray. Count and write how many of each kind.

Answer: 4 shells; 5 buttons; 3 spoons

Count one kind at a time and cross out each thing. 4 shells, 5 buttons and 3 spoons. 4 + 5 + 3 = 12, the same as the things on the tray, so no thing is missed.

Watch for: Counting a thing twice.; Missing a thing in the middle of the tray.

S2. The tally shows how many things Ali counted.

Answer: 6 pens; 8 rubbers

The first group is 5. Pens: 5 and 1 more is 6. Rubbers: 5 and 3 more is 8.

Watch for: Counting a group of 5 as 4.; Counting each group as 1.

S3. Look at the graph. Each picture stands for 1 pet.

Answer: 3 more cats than fish; 17 pets in all

Cats 7 and fish 4. 7 take away 4 is 3 more cats. All three rows: 7 + 4 + 6 = 17 pets.

Watch for: Adding 7 and 4 to find how many more.; Leaving out a row when finding the total.

S4. Ali ate 2 apples, 5 pears and 3 plums. Draw one picture in a box for each fruit.

Answer: Apples 2 pictures; pears 5 pictures; plums 3 pictures

Draw 2 pictures in the apples row, 5 in the pears row and 3 in the plums row. Some boxes are left empty. The three rows have 2 + 5 + 3 = 10 pictures, one for each fruit.

Watch for: Filling every box in a row.; Drawing the pictures in different sizes.

S5. Look at the graph. Each picture stands for 1 child.

Answer: Chess; 13 children

Chess has 3 pictures. That is the shortest row, so the fewest children play chess. Ludo 5 and cards 8: 5 + 8 = 13 children.

Watch for: Choosing the longest row.; Adding the chess row as well.

S6. Look at the graph. Each picture stands for 1 stamp.

Answer: True, False, True, True

Ali 5, Mei 5 and Ben 2. Ali and Mei have the same number: true. Ben has the fewest, not the most: false. 5 take away 2 is 3, so Mei has 3 more than Ben: true. 5 + 5 + 2 = 12: true.

Watch for: Saying Ben has the most because his name is last.; Forgetting to add all three rows.

Think harder (page 3)

T1. The tally shows the things in a box. Fill in the table.

Answer: 9, 5, 6; 20

Hats: 5 and 4 more is 9. Bags: 5. Shoes: 5 and 1 more is 6. 9 + 5 + 6 = 20 things in all.

Watch for: Counting a group of 5 as 4.; Adding only two of the kinds.

T2. Study the graph. Each picture stands for 1 thing. Which question can the graph answer? Write its number in the brackets.

Answer: (2)

The graph shows how many of each kind are in the box. Question (2) asks about the number of shells and stones: 6 take away 2 is 4 more shells. The other questions are about likes, places or looks, and the graph does not show them.

Watch for: Choosing (1) because it sounds like a graph question.; Choosing a question about the prettiest.

T3. There are 16 books in all. Draw the art books in the empty boxes.

Answer: 11 maths and story books; 5 art books

Maths 4 and story 7 make 11. 16 take away 11 is 5. Draw 5 pictures in the art row. Check: 4 + 7 + 5 = 16.

Watch for: Writing 16 or 7 for the art books.; Drawing a picture in every empty box.

Go further (page 4)

G1. Mei's graph uses a key. Each picture stands for 2 sweets. Draw the pictures for the toffees in the empty boxes.

Answer: 3 pictures; 7 gums

6 toffees: 2, 4, 6 is 3 pictures. The gums row has 3 whole pictures, 6 sweets, and half a picture, 1 more: 7 gums.

Watch for: Drawing 6 pictures for 6 toffees.; Counting half a picture as 2 or as 0.

G2. Four children have 21 stickers in all. Ben has 4 stickers. Lily has 3 more than Ben. Jamie has 2 fewer than Lily. Ravi has the rest.

Answer: Lily 7; Ravi 5

Lily: 4 and 3 more is 7. Jamie: 7 take away 2 is 5. Ben, Lily and Jamie have 4 + 7 + 5 = 16. 21 take away 16 is 5, so Ravi has 5. Check: 4 + 7 + 5 + 5 = 21.

Watch for: Writing 3 for Lily (the number more, not the total).; Forgetting to take Ben away when finding the rest.

G3. Study the graph. Write a question that the graph can answer. Then write the answer. Many answers are right.

Answer: Many answers are right

Any question about the numbers on the graph is good, with the right answer. For example: How many more tulips than daisies? 6 more. How many flowers are there in all? 17. Which flowers have the fewest? Daisies. A question about which flower a child likes cannot be answered.

Watch for: Writing a question the graph cannot answer, such as which flower is nicest.; Writing a question with no answer.

Investigation (page 5): Sort, draw and ask

Many answers are right. A good graph has a title, the kinds named, one picture for each thing, all the pictures the same size, and a key. The numbers on the graph match the piles.

Look for: A child who sorts by one feature, draws one picture for each thing and writes a question the graph can answer (how many, how many more, which has the fewest). Credit a clear tally. A child who writes a question about likes has not yet met the idea: ask what the graph shows. A child who draws a graph with a key of 2 has gone further.

Exit ticket (page 5)

E1. Look at the tray.

Answer: 3 rubbers

Cross out each rubber as you count it: 3 rubbers. The other 4 things are pencils. 4 + 3 = 7, the same as the things on the tray.

Watch for: Counting a thing twice.; Counting the pencils as well.

E2. Look at the graph. Each picture stands for 1 biscuit.

Answer: 5 more biscuits

Ali 7 and Mei 2. 7 take away 2 is 5 more biscuits.

Watch for: Adding 7 and 2.; Writing 7.

E3. A child counts the toys in a box and draws a graph. Which question can the graph answer? Write its number in the brackets.

Answer: (1)

The graph shows how many toys of each kind are in the box, so question (1) can be answered. The graph does not say which toy is the best.

Watch for: Choosing (2): the graph does not show what a child likes.

Sneak-peek puzzles (not in the workbook)

1. 4 spoons.
2. Any clear picture: for example a row of 4 spoons, a row of 3 buttons and a row of 2 shells, one small picture for each thing.